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Efficient multi-agent deep reinforcement learning algorithm for multi UAV collision avoidance

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HIGHLIGHTS

- Suggest an efficient multi-agent deep reinforcement learning algorithm for UAV collision avoidance in dynamic environments.
- Propose an efficient graph attention network architecture for modeling UAV interactions.
- Demonstrate improved scalability with increasing numbers of UAVs based on the suggested architecture and curriculum learning.

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ABSTRACT

The rapid expansion of unmanned aerial vehicles (UAVs) across industries has led to increased airspace congestion. The increasing use of drones across many fields and locations has caused serious problems, especially in avoiding collisions. In the rapidly developing field of drone technology, ensuring UAV flight safety and reducing the risk of UAV collisions have therefore become urgent concerns. There are many artificial intelligence (AI) algorithms designed to solve this problem, but most work only in situations with a single agent. Multi-agent reinforcement learning is a promising way to solve these problems. It enables drones to operate with greater intelligence and flexibility, even in challenging situations, alongside other agents. This work presents a Multi-Agent Deep Reinforcement Learning algorithm based on efficient graph attention network for collision avoidance in a dense, complex multi UAV environment. We propose both curriculum learning and transfer learning by adding more agents over time and subsequently employing learning models. This makes the system more scalable and more coordinated. The training process is significantly enhanced by the suggested method, which outperforms the current baselines in continuous settings. Our findings indicate that the proposed approach achieves 17% higher cumulative reward, up to 10% fewer loss-of-separation time steps, and about 44% fewer active interaction edges than the benchmark. Furthermore, the proposed method reduces action-selection bias, improving decision-making stability in dense multi-UAV settings.

1. Introduction

The advent of autonomous vehicles signifies an excellent opportunity for improving safety and maximizing energy and fuel-minimal use in traffic systems, rendering them an essential component of future sustainable transportation networks [1]. The public mainly focuses on self-driving cars [2]; nevertheless, autonomous vehicles' advantages include various transportation modalities, including autonomous watercraft [3] and aviation systems [4]. Irrespective of whether a vehicle is operated by a human or autonomously, the primary objective in all traffic operations is to ensure safety by averting accidents with other vehicles and static obstructions [5,6].

In the following years, considerable growth in the worldwide market for unmanned aircraft systems (UAS), both commercial and civil, is anticipated. The Single European Sky Air Traffic Management Research (SESAR) predicts that by 2035, the European drone industry will have grown to over 10 billion euros annually. By 2050, it will reach 15 billion euros. The mission characteristics and application sectors will determine

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which UAS are expected to provide the best market value: small UAS and low-level airspace activities. This expansion likely results in more traffic jams and problems with efficiency, dependability, and safety. Thus, integrating UAS into civil airspace requires creating and using conflict management systems [7].

The Federal Aviation Administration (FAA) emphasizes the importance of developing a collision-free route creation and navigation system for UAVs. Ensuring safe, risk-free UAV flights is equally important for indoor use, where stability is a key priority. This requires UAVs to detect and avoid both stationary and moving objects, such as buildings and trees (static obstacles) or birds (dynamic obstacles) [8].

The goal of artificial intelligence (AI) is to make robots as intelligent as humans. AI in computer science is the study of intelligent agents, or anything that can sense the environment and behave to increase its chances of accomplishing certain objectives [9,10]. The rapid spread of drone use in many different fields and settings has highlighted how urgently efficient collision avoidance systems are needed. Though many artificial intelligence (AI) algorithms and conventional techniques have been tried to solve this problem, their usefulness is often restricted to single-agent systems and they do not scale well to multi-agent settings.

Reinforcement learning (RL) is a branch of machine learning where an agent learns to make decisions by interacting with its surroundings and aiming to maximize rewards. The policy that governs the agent's activities, the reward system that the environment provides in response to those actions, and the models that help the agent understand its surroundings are the core components of reinforcement learning [9,12,13]. Combining reinforcement learning with deep learning makes it more effective, capable of handling complex decisions over large state spaces and sometimes outperforming human judgment [14]. However, there are limitations with the reinforcement learning-based conflict resolution techniques currently in use, such as in air traffic control [15–17]. When dealing with increasingly complex, high-density scenarios, especially in growing cities [11] that call for agent collaboration and interaction, these methods struggle since they often focus on the individual agents [7,17].

Even though learning-based airspace deconfliction has made significant progress, there is still a clear scalability gap for dense multi-UAV settings. Isufaj et al. [17] suggest using graph reinforcement learning to solve multi-UAV conflicts. However, neighborhood aggregation may become less selective as the swarm grows and the interaction graph becomes dense, which can increase the amount of work needed to pass messages and mix up safety-critical neighbors with ones that aren’t relevant. From the point of view of air traffic control, Papadopoulos et al. [18] show how deep reinforcement learning can help with tactical conflict resolution. However, these ideas are more related to centralized decision support and don’t really address decentralized multi-agent coordination in rapidly changing local conflict environments. Huang et al. [19] use counterfactual credit assignment to help multiple UAVs avoid collisions. However, dense interactions and non-stationarity still cause problems when there are more agents and when the communication and interaction structure is noisy or all-to-all. In the meantime, recent studies using graph-based MARL are still improving the capabilities of UAV swarms by using learned graph embeddings and attention [38,39]. However, many of these studies focus on tasks like avoiding or tracking obstacles instead of separating UAVs in shared airspace, and they don’t make it clear that they are using a conflict-driven, edge-pruned interaction topology along with curriculum-to-transfer scaling. Recent graph-based MARL works [38,39] are still improving UAV-swarm abilities with learned graph embeddings and attention. Nevertheless, many of them are more concerned with issues like tracking or avoiding objects than with tactical separation in shared airspace.

We use the attention mechanism to prioritize important UAVs that need a quick reaction by dynamically assessing the spatial connections and possible collision paths between many drones. Through continuous operation, this real-time analysis allows the system to undertake collision avoidance maneuvers that are more accurate and timelier than those of traditional standards.

The strategy is validated for resilience and scalability through a long sequence of controlled simulations that mimic various real-world situations. The collision avoidance skills are shown to be significantly improved by these simulations, where our method reduces incident rates and enhances overall safety and coordination among numerous drones.

By improving the drone training procedure, this study also advances the sector. More subtle learning is made possible by the incorporation of sophisticated attention processes into graph networks, which enable drones to swiftly and successfully adapt to complex surroundings. This development increases drone fleet operational safety and extends their functional lifespan and reliability in dynamic and unpredictable environments.

The proposed attention graph network algorithm represents a significant leap forward in multi-agent drone collision avoidance. The empirical evidence provided by our simulations highlights its potential to become a cornerstone technology in the safe and efficient deployment of drones across various sectors, paving the way for more sophisticated and resilient drone applications. We propose IGAT-MARL, a conflict-resolution approach that couples conflict-driven interaction modeling with an improved graph-attention value network and curriculum/transfer training. Unlike other distance-threshold interaction graphs that can become dense as the swarm size increases [17,19,38], our approach prioritizes only safety-critical neighbors and learns more selective coordination policies in dense multi-UAV settings.

We make the following contributions in this work:

1. We introduce a conflict-aware interaction modeling strategy that updates inter-UAV connectivity online, enabling UAVs to focus coordination on safety-relevant neighbors while limiting unnecessary interactions as swarm density increases. We further apply conflict-gated execution so that heading commands are applied only to UAVs currently involved in conflicts.
2. We propose an Improved Graph Attention backbone (IGAT), tailored for dense UAV conflict resolution by combining multi-head GAT energy scoring with residual connections and layer normalization within a stacked (multi-pass) message aggregation pipeline, enabling robust information fusion under sparse, time-varying graphs.
3. We present a curriculum learning schedule that progressively increases UAV numbers (3 → 10 UAVs), transferring network parameters between stages to stabilize learning under non-stationarity and accelerate convergence.
4. We validate IGAT-MARL in BlueSky simulator using aircraft performance constraints (BADA) and evaluate scalability using four safety metrics (reward, loss of separation, action bias, and number of edges), demonstrating improved coordination under dense conflict scenarios.

The paper is laid out as follows. The second part discusses similar efforts on UAV collision avoidance systems utilising Reinforcement Learning . In the third part, we discuss our suggested IGAT Multi-agent Reinforcement Learning approach. The fourth part analyses the results, and the fifth part discusses the research results and outcomes. Finally, the report concludes with a conclusion and suggestions for further research.

2. Related works

Using multi-agent learning techniques has made a significant difference in the field of UAV collision avoidance. The implementation of Multi-Agent Reinforcement Learning (MARL) has demonstrated potential to facilitate UAV swarm navigation through complex environments while avoiding collisions. Huang et al. [19] presented MACA, a multi-agent critic-actor learning system that considers energy efficiency and safety while optimizing a global reward through a centralized critic.

Xu et al. [20] used the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) method to address route planning and target allocation problems in multi-UAV systems. Their results showed that deep learning methods could help overcome significant navigation challenges. To achieve scalable and reliable multi-UAV route planning without UAV-to-UAV interaction, Huang et al. [21] provide a vision-based decentralized collision avoidance policy learning technique using deep reinforcement learning (DRL), which primarily leverages onboard vision sensors and inertial measurements. Ahmad Taher Azar et al. [22] focuses on reinforcement learning (RL) approaches and their performance in this situation, which incorporates the issues of UAV control, navigation, and route planning. A Markov Decision Process (MDP) framework is used to analyze mobility and multi-UAV edge computing failures. Another study [23] introduces a novel distributed deep reinforcement learning (DRL) technique that utilises joint exploration and experience replay priority. Additionally, Seong et al. [24] provide a MARL technique. To enable wireless sensor networks, autonomous UAVs may aid with route planning, charging capacity, and other needs, including collision–avoidance. Moreover, Yun et al. [16] study air-to-ground ultra-reliable and low-latency communication, combining graph attention exchange networks and multi-agent DRL to handle UAV coordination while chasing a moving ground user.

The Hart et al. [5] work integrates Reinforcement Learning (RL) with a supervised learning strategy to assess collision risk parameters. This makes the agent better at handling tasks that require avoiding moving obstacles. D'Apolito and Sulzbachner [25] used reinforcement learning techniques to train agents to prevent mid-air collisions with uncooperative intruders.

Reinforcement learning (RL) research has made significant progress, particularly in the application of RL-based solutions to resolve real-world collisions. The RL method generally has two main phases. The first stage is to train the agent (a neural network) to learn a good policy in a given RL environment. The next stage is to employ the trained agents to handle real-life collisions. The training phase can take a while, but the solution phase moves quickly, enabling excellent results. RL techniques are ideal for obtaining quick responses in operational environments because they handle much of the arithmetic during training. The Markov decision process (MDP) is the foundation of RL and helps people make decisions [26].

The evolving integration of UAVs across numerous sectors has led to a large growth in UAV usage, it requires appropriate management to avoid risks like collisions. Multi-agent reinforcement learning (MARL) algorithms, which are vital to improving drone collision avoidance, significantly depend on the quality of agent interaction. This interaction is governed by network topologies that allow low latency, dependability, and efficiency are critical characteristics for effective agent communication. The capacity of agents to perceive their environment has considerably expanded due to recent breakthroughs in Graph Convolutional Reinforcement Learning, allowing more effective decision-making in dynamic situations. The scalability and number of UAVs, communication networks, the performance of MARL, the intricacy of multi-agent interactions in real-world settings, and the integration of these systems under extremely dynamic conditions are obstacles that restrict their wider application and real-time implementation, need more research [7].

2.1. MARL-based collision avoidance for UAV systems

Multi-agent reinforcement learning (MARL) has become a widely used paradigm for decentralized collision avoidance (CA) in UAV swarms, since it can learn adaptive policies directly from interaction experience without requiring an explicit analytic model of the environment [40–43]. Early work by Wang et al. [40] proposed a two-stage RL approach under imperfect sensing, showing that staged learning and reward structuring can improve convergence under partial observability. However, such approaches often rely on limited interaction modeling and may not capture rich relational structure among many simultaneously interacting UAVs, can restrict scalability as swarm density increases.

2.2. Optimization-inspired RL and cross-domain multi-agent coordination

Optimization-inspired reinforcement learning has been studied for large-scale and constrained decision-making problems that share practical challenges with dense multi-UAV coordination (e.g., high-dimensionality and competing objectives). Ma et al. [44] introduced reference-vector reinforcement learning for constrained many-objective optimization in an industrial setting, and an adaptive localized decision-variable analysis to reduce search complexity in large-scale multi-objective optimization [45].

Communication-aided designs have also used tiered training, in which agents first learn how to avoid problems on their own and then how to work together [43]. The cooperative, decentralized collision avoidance algorithm CoDe [42] further demonstrated the importance of credit assignment in continuous control, but its evaluation scope remained confined to modest-sized teams. Multi-agent DRL has also been used for cooperative vision and computation offloading, necessitating agent coordination within the constraints of communication and resource limitations. Zhao et al. [46] looked at multi-agent DRL for co-operative sensing and computation in vehicular edge computing. They showed how learned cooperation may make a system work better when there are decentralized limitations. Hybrid learning techniques have also been investigated for dynamic transportation networks and routing challenges [47,48].

Finally, MARL has also been extensively used for *other* multi-UAV coordination problems, such as resource scheduling and task allocation in UAV-assisted wireless networking [49–51]. These studies reinforce that decentralized MARL can coordinate heterogeneous agents under dynamic conditions, but they focus on communication or resource management rather than collision avoidance, and typically do not address conflict-centric interaction graphs for safe navigation.

2.3. Graph neural networks and attention mechanisms in multi-UAV MARL

Graph neural networks (GNNs) and attention mechanisms provide natural inductive biases for modeling inter-agent relationships in multi-UAV systems, where coordination quality depends strongly on which neighbors influence each decision. Recent work has incorporated graph-based interaction modeling into UAV collision avoidance and related safety settings [17,19,34,35,38]. While graph representations can reduce the burden of fully connected communication, many commonly used constructions (e.g., static or distance-threshold graphs) can become dense as N increases, mixing safety-critical interactions with incidental ones and degrading scalability.

Attention mechanisms can mitigate this issue by learning to weight neighbors instead of treating all admitted edges equally. Graph-attention critics and relational attention modules have been reported to improve selectivity in multi-agent settings by focusing computation on the most relevant entities in the neighborhood [52,53]. Transformer-based MARL has also been explored for variable-size neighborhood processing and coordination [54]. However, transformer-style models can be computationally heavy for dense swarms, and evaluations are often conducted in structured settings that may not fully reflect the interaction noise found in high-density multi-agent collision-avoidance scenarios.

More broadly, GNN-empowered MARL has been applied in multi-agent networking and resource coordination, further underscoring the general utility of graph-structured reasoning for decentralized decision making [46,55]. These settings differ from collision avoidance, but they support the recurring insight that explicit relational modeling improves scalability.

In graph learning, multi-head Graph Attention Networks (GATs) provide a strong baseline for neighbor weighting [27]. However, in dense multi-agent settings, standard multi-head GAT message passing can still suffer when the underlying graph becomes overly connected or when attention is forced to operate over many weakly relevant neighbors.

Some recent research has used graph reinforcement learning (GRL) to address conflicts by explicitly modeling graph interactions among cars and developing rules using graph embeddings. Li et al. [37] propose that a GRL framework for resolving multi-aircraft conflicts in time-varying conflict graphs, using engineered interaction weights and graph-based value approximation, is advantageous for improving coordination under organized airspace limits. Also, Li et al. [36] suggest a graph-scaled RL chain that combines local and global conflict-risk elements, such as graph-based feature refinement and action-space safety goals, so that it can work in very crowded situations and face very tough difficulties.

2.4. Curriculum learning and transfer in multi-agent training

Curriculum learning (CL) teaches agents to perform tasks that become harder over time. It has been proven multiple times to make optimization more stable in complicated reinforcement learning problems [56]. In multi-agent training, curricula are particularly beneficial since dense interactions may result in unstable value estimates and prolonged convergence. In the field of UAVs, curriculum strategies that gradually increase scenario complexity (including intruder count, barriers, or interaction density) have demonstrated superior convergence and final policy quality compared to direct training on the most challenging distribution [41,43,54]. People have also explored using curriculum-based training to enhance stability in more complex settings. Yan et al. [41] proposed a task-specific curriculum for fixed-wing UAVs to prevent collisions when flocking in environments with numerous obstacles. They found that this approach made learning more stable and reduced the number of collisions compared to non-curricular baselines. It is also common to apply transfer learning across stages by initializing later stages with earlier parameters to maintain training stability as it becomes more complex [57].

When looked at as a whole, current MARL-based research on avoiding collisions shows two persistent gaps for dense UAV swarms. First, when N becomes bigger, many interaction graphs are noisy or too crowded (particularly with static or distance-based neighborhoods). This may lead to interactions that don't matter and worsen coordination. Second, value learning is still hard when scaling from small to large swarms, even when curriculum learning is applied.

3. Methodology

We examine a crowded multi-UAV collision-avoidance scenario simulated in the BlueSky air-traffic simulator [29]. Let N be the number of UAVs (agents). At decision step t, UAV $i \in \{1, \dots, N\}$ is characterized by latitude ϕ_i^t, longitude λ_i^t, heading ψ_i^t (degrees), calibrated airspeed cas_i^t, and altitude h_i^t. The simulator performs state transitions and computes conflict metrics—the MARL controller communicates with the simulator by sending discrete heading instructions. Every K steps, the MARL policy is put into action. Let Δt_{sim} be the time step and $\Delta t = K\Delta t_{\mathrm{sim}}$ be the time between MARL decisions. The regulated state transition may be expressed abstractly as

$$s^{t+1} \sim P(\cdot \mid s^t, \mathbf{a}^t), \tag{1}$$

where s^t is the state and $\mathbf{a}^t = (a_1^t, \dots, a_N^t)$ are the agents' actions. This abstraction corresponds to the implementation of $(\phi_i^t, \lambda_i^t, \psi_i^t, \mathrm{cas}_i^t, h_i^t)$ and the policy issues directing instructions.

A protected-zone radius (RPZ) sets the minimum distance between two points on the horizontal plane. BlueSky provides information on conflicts based on look-ahead conflict detection and closest-point-of-approach measurements. The simulator provides the distance at the closest point of approach DCPA_{ij}^t and the time to CPA TCPA_{ij}^t for each pair of UAVs (i, j). If $\mathrm{DCPA}_{ij}^t < r_{\mathrm{pz}}$ within the given look-ahead horizon, there is a possibility of a conflict. An occurrence termed a loss-of-separation (LOS) happens when the instantaneous separation drops below the protected-zone level.

Each UAV receives a local observation vector that aligns with the implementation:

$$o_i^t = \left[\frac{\phi_i^t}{100}, \frac{\lambda_i^t}{100}, \frac{\psi_i^t}{180} - 1, \mathrm{Mach}(\mathrm{cas}_i^t, h_i^t) \right]. \tag{2}$$

Each UAV selects a discrete action

$$a_i^t \in \{0, 1, 2\}, \tag{3}$$

mapped to commanded heading offsets:

$$\Delta\psi(a_i^t) = \begin{cases} 0^\circ, & a_i^t = 0, \\ +15^\circ, & a_i^t = 1, \\ -15^\circ, & a_i^t = 2. \end{cases} \tag{4}$$

In the implementation, heading instructions are only sent to UAVs that are currently engaged in conflicts. If UAV i is not involved in a conflict at time t, its heading command does not change (conflict-gated execution).

Dynamic Graph Construction: The exchanges between agents are shown as a graph that changes over time, with $G^t = (V, E^t)$, Nodes are UAVs, and edges show how they interact with each other in ways that are important for safety (conflict). Let C^t denote the set of conflict pairs at time t. We define

$$E^t = \{(i, j) \mid (i, j) \in C^t\}, \tag{5}$$

and the adjacency matrix $A^t \in \{0, 1\}^{N\times N}$ as

$$A_{ij}^t = \begin{cases} 1, & (i, j) \in E^t, \\ 0, & \text{otherwise}, \end{cases} \qquad A_{ii}^t = 0. \tag{6}$$

At each decision step t, C^t changes and the environment rebuilds A^t directly from the current conflict pairs. This means that edges only show up when conflicts are found and go away when they are resolved, creating a sparse interaction graph that changes over time. Case in instance: if $N = 5$, $C^t = \{(1, 3), (2, 5)\}$ then

$$A^t = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \tag{7}$$

and if at the next step $C^{t+1} = \{(1, 2)\}$ then

$$A^{t+1} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{8}$$

3.1. Mathematical formulations

Reinforcement learning defines a Markov Decision Process (MDP) [28] with state space $\mathcal{S}$, action space $\mathcal{A}$, transition P, reward R, and discount $\gamma \in (0, 1)$. In the multi-UAV setting, we learn decentralized policies through per-agent action-value functions while interacting with the shared state.

An Improved Graph Attention Network (IGAT) is used in the training model for multi-agent settings. In these settings, interactions are represented as a graph of nodes and edges that show how agents communicate. Two GAT layers are stacked in each block, and there are two blocks in the whole network. Each GAT layer uses multi-head attention with residual links and normalization.

Let $\mathbf{H} \in \mathbb{R}^{N\times d_{\text{in}}}$ represent the initial node features, and let $A^t \in \{0,1\}^{N\times N}$ represent the adjacency matrix from (7). Define $\mathcal{N}(i) = \{j \mid A^t_{ij} = 1\}$.

Linear transformation: Each node feature h_i is transformed using a learnable weight matrix $W \in \mathbb{R}^{d_{\text{out}}\times d_{\text{in}}}$:

$$h_i^{(W)} = W h_i. \tag{9}$$

Pairwise attention energy: For each node pair (i,j), a GAT-style energy is computed [27]:

$$e_{ij} = \text{LeakyReLU}\left(\mathbf{a}^T \left[h_i^{(W)} \parallel h_j^{(W)}\right]\right), \tag{10}$$

where $\mathbf{a} \in \mathbb{R}^{2d_{\text{out}}}$ is learnable and $\parallel$ denotes concatenation.

Masked attention: Non-neighboring nodes are masked using A^t:

$$\hat{e}_{ij} = \begin{cases} e_{ij}, & A^t_{ij} = 1, \\ -\infty, & \text{otherwise.} \end{cases} \tag{11}$$

Softmax normalization:

$$\alpha_{ij} = \frac{\exp(\hat{e}_{ij})}{\sum_{k\in\mathcal{N}(i)} \exp(\hat{e}_{ik})}. \tag{12}$$

Node representation update:

$$h'_i = \sum_{j\in\mathcal{N}(i)} \alpha_{ij}\, h_j^{(W)}. \tag{13}$$

Multi-head attention: With K attention heads, each head computes $h_i^{\prime(k)}$, and outputs are concatenated:

$$h'_i = \big\|_{k=1}^{K} h_i^{\prime(k)}. \tag{14}$$

Residual connection and layer normalization: Each attention layer uses a residual connection followed by LayerNorm:

$$h_i^{\text{IGAT}} = \text{LayerNorm}\left(h'_i + \text{Res}(h_i)\right). \tag{15}$$

Q-value estimation: The Q-network maps each UAV embedding to action-value predictions:

$$\mathbf{Q}_i(o_i^t) = W_q\, h_i^{\text{IGAT}} + b_q, \qquad W_q \in \mathbb{R}^{|\mathcal{A}|\times d_h}. \tag{16}$$

3.2. Problem formulation

We model multi-UAV collision avoidance as a decentralized partially observable multi-agent decision process [58]. Let π_θ denote the decentralized policy induced by learned action-value functions. The learning objective is to maximize the expected discounted team return:

$$\max_\theta\ J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T} \gamma^t \left(\frac{1}{N}\sum_{i=1}^{N} r_i^t\right)\right]. \tag{17}$$

3.2.1. Reward design

The reward is based on three factors: the number of present conflicts, off-track movement measured by deviation from a reference or closest-conflict direction, and a CPA-based risk term calculated from the DCPA^t_{ij} relative to r_{pz}:

$$r_i^t = -\underbrace{\frac{|\psi_i^t - \psi_{i,\text{ref}}^t|}{\psi_{\max}}}_{\text{off-track}} - \underbrace{n_i^t}_{\text{conflict count}} - \underbrace{\max_{(i,j)\in C^t}\left(1 - \exp\left(1 - \frac{1}{\sqrt{\max(x_{ij}^t, \epsilon)}}\right)\right)}_{\text{CPA risk}}, \tag{18}$$

where n_i^t is the number of conflicts involving UAV i at time t, $x_{ij}^t = \text{DCPA}^t_{ij}/r_{\text{pz}}$, ϵ is a small constant for numerical stability, and $\psi_{\max}$ is a normalization constant.

3.3. Proposed IGAT-MARL

The proposed IGAT-MARL in Fig. 1 uses the value-function approximator over the dynamic conflict graph A^t. At each time step, an encoder transforms the local observation from each UAV into a hidden node embedding $\mathbf{H}^t \in \mathbb{R}^{N\times d}$. Then, stacked graph-attention blocks handle coordination between agents. These blocks only send messages along the edges of A^t. They do not send messages over a fully connected graph. When relationships between agents change over time, LayerNorm and residual links keep learning stable. A linear head generates Q-values for each agent. The term "double attention" refers to two types of repeated attention passes: two attention layers in a row within each IGAT block ($\text{GAT}_1 \to \text{LN} \to \text{GAT}_2 \to \text{LN}$); and stacking two IGAT blocks in the architecture as a whole, which allows multiple refinement passes over A^t as conflicts emerge and resolve. To make it easier to scale up and reduce non-stationarity in dense swarms, training uses a curriculum approach that gradually adds more UAVs, while transfer learning starts each stage with the parameters learned in the previous stage.

Algorithm 1 shows the overall multi-UAV collision avoidance training loop using off-policy DQN updates and the proposed IGAT value network.

4. Simulations and results

The Air Traffic Simulator BlueSky was used to model and study UAV behavior [29]. Additionally, to make the scenario more realistic, we used the BADA dataset from EuroControl [30], which includes small planes that closely resemble fixed-wing UAVs. All models ran smoothly on an Ubuntu computer with an NVIDIA A40 GPU, which had enough processing power for high-resolution modeling and sound data processing. For 10,000 different episodes, the models were run with 3 to 10 UAVs.

4.1. Performance metrics

In the assessment of UAV collision avoidance systems based on multi-agent learning, a range of critical metrics are used. Cumulative rewards illustrate how effectively the system functions over time by ensuring that objectives are met without any issues. The Total Loss of Separation is an important safety metric since it indicates how quickly and closely UAVs may approach harmful distances. Action Bias examines whether the system is more biased toward specific actions than others, or whether it makes judgments fairly to ensure consistent, equitable performance. The Number of Edges represents the number of interactions or connections between UAVs. This is the final method to determine how complex the system and agent communication are in avoiding a crash. These factors help us understand how flexible, efficient, and safe a UAV system is.

4.2. Simulation results

The baseline we utilised for our comparison is more accurately defined as a DGN module based on attentiveness. It uses Q/K/V projections and masked dot-product attention across neighbors, and it uses the binary adjacency/mask A^t to allow or stop message passing. In contrast to a traditional GCN layer, which combines neighbor characteristics using a normalized adjacency without Q/K/V attention, this one does. Both IGAT-MARL and the benchmark use the simulator's conflict pairs to construct a time-varying binary adjacency matrix A^t, but the way the graph is used within the network differs. The benchmark DGN(QKV) mainly employs A^t as a hard feasibility mask in a dot-product attention module. A single-pass masked softmax over $QK^\top$ weighs neighbor influence, while A^t decides if communication happens. On the other hand, IGAT-MARL develops a sparse, conflict-driven topology and then learns edge-wise significance using a GAT-style pairwise energy, followed by normalization to obtain attention coefficients solely on active edges. This provides a useful weighted adjacency, $\tilde{A}^t_{ij} \triangleq A^t_{ij}\cdot\alpha^t_{ij}$, that we utilise to figure out how edges impact each other. This changes as conflicts arise or resolve (via A^t) and as learned embeddings change (through

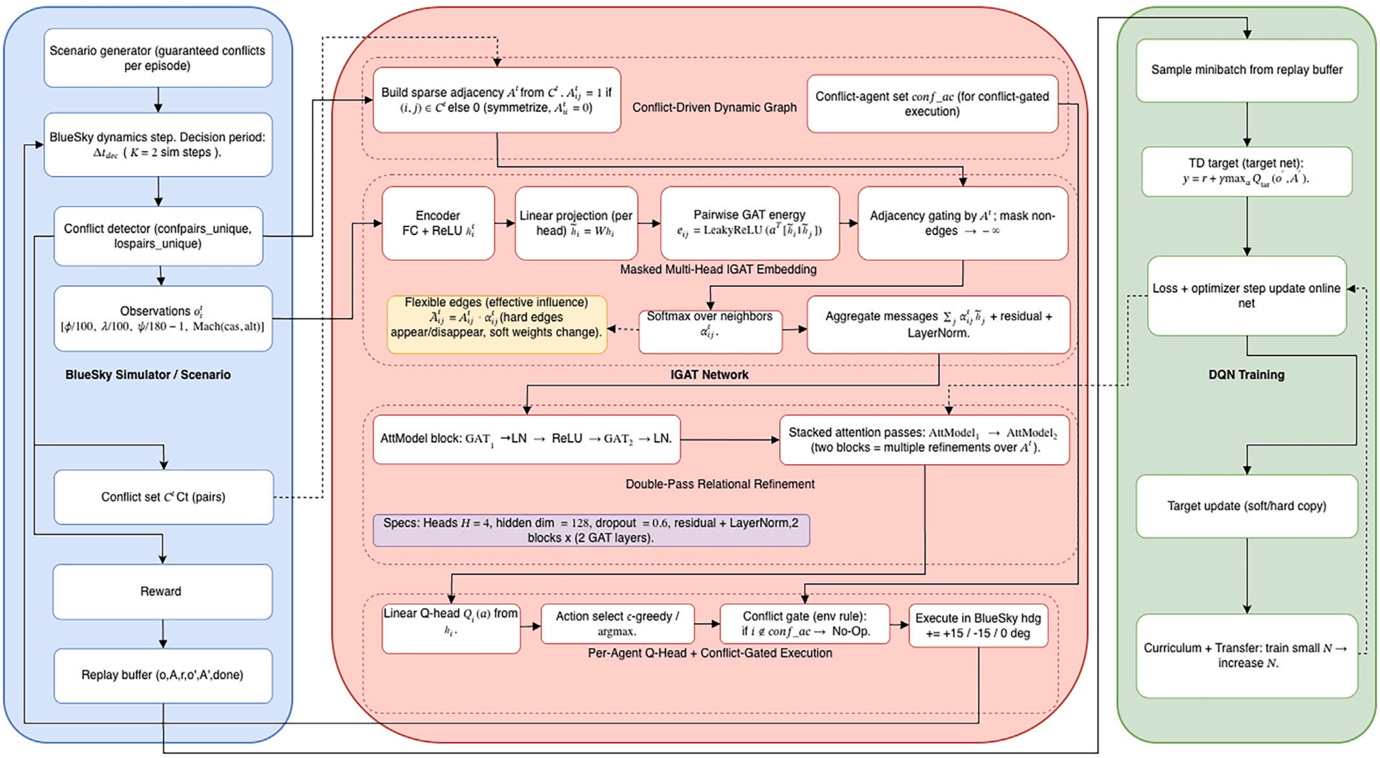


Fig. 1. IGAT-MARL framework for UAV collision avoidance.

Algorithm 1 Efficient Multi-Agent Reinforcement Learning Collision Avoidance Algorithm based on IGAT.

Initialize environment, online network, target network, optimizer, replay buffer
for each episode **do**
  Reset environment; obtain initial observations and adjacency A^t
  while episode not done **do**
    Compute per-agent Q-values using IGAT and A^t
    Select actions via ϵ-greedy
    Apply heading commands (conflict-gated execution)
    Step simulator; receive next observations, reward, and updated adjacency A^{t+1}
    Store transition $(o^t, a^t, r^t, o^{t+1}, A^t, A^{t+1})$ in replay buffer
    Sample minibatch and update online network using TD loss
    Periodically update target network
  end while
  Log episode statistics
end for

α^t). In addition, IGAT improves these edge weights by using stacked multi-head GAT layers with residual connections and layer normalization (known as "double attention"). This creates a more selective and stable conflict-centric interaction structure than a masked dot-product attention baseline that works under the same simulator constraints.

4.2.1. Cumulative rewards

Fig. 2 illustrates the total reward over training for swarm sizes from N=3 to N=10. In all settings, IGAT shows a clearer learning trend and achieves a larger (less negative) reward than the benchmark. This demonstrates that the quality of the policy improves as more experience is gained. The gap becomes more noticeable as N increases, which suggests that IGAT maintains more effective coordination under higher interaction density. Overall, the reward curves support that IGAT enhances both convergence behavior and final performance compared to the baseline graph-value module [17].

4.2.2. Total loss of separation

Safety is assessed by the cumulative duration spent in loss-of-separation (LoS) circumstances, with reduced values signifying expedited dispute resolution and a reduction in extended separation infractions. Fig. 3 illustrates that IGAT always lowers t_{loss} compared to the benchmark, even as the setting becomes more crowded (higher N). This indicates that IGAT improves conflict handling under dense multi-UAV interactions, where prolonged LoS is more likely to compound into unsafe events.

4.2.3. Agents actions and action bias

Action selection bias is important in RL-based collision avoidance because persistent preference toward a single maneuver (e.g., repeatedly choosing 1 action or always turning in one direction) can reduce responsiveness and lead to slower conflict resolution in unseen configurations [31–33]. Fig. 4 illustrates representative action distributions. IGAT creates a more flexible action profile than the benchmark by employing turns as necessary and without letting one action dominate excessively. This behavior corresponds with enhanced rewards and reduced t_{loss}, indicating better adaptive decision-making amidst fluctuating conflict patterns.

4.2.4. Graph sparsity and number of active edges

The edge count reflects the *graph complexity* induced by the conflict-driven adjacency A^t, i.e., how many pairwise interactions are active on average. Importantly, fewer edges should not be interpreted as "fewer collision areas" by itself; rather, it indicates that the learned coordination can operate with a *sparser interaction structure*, which reduces unnecessary message passing and computational overhead. Table 1 indicates that IGAT has fewer active edges than the benchmark for all agent counts, particularly in denser situations. This, together with the reward and LoS findings, implies that IGAT achieves better safety-efficiency

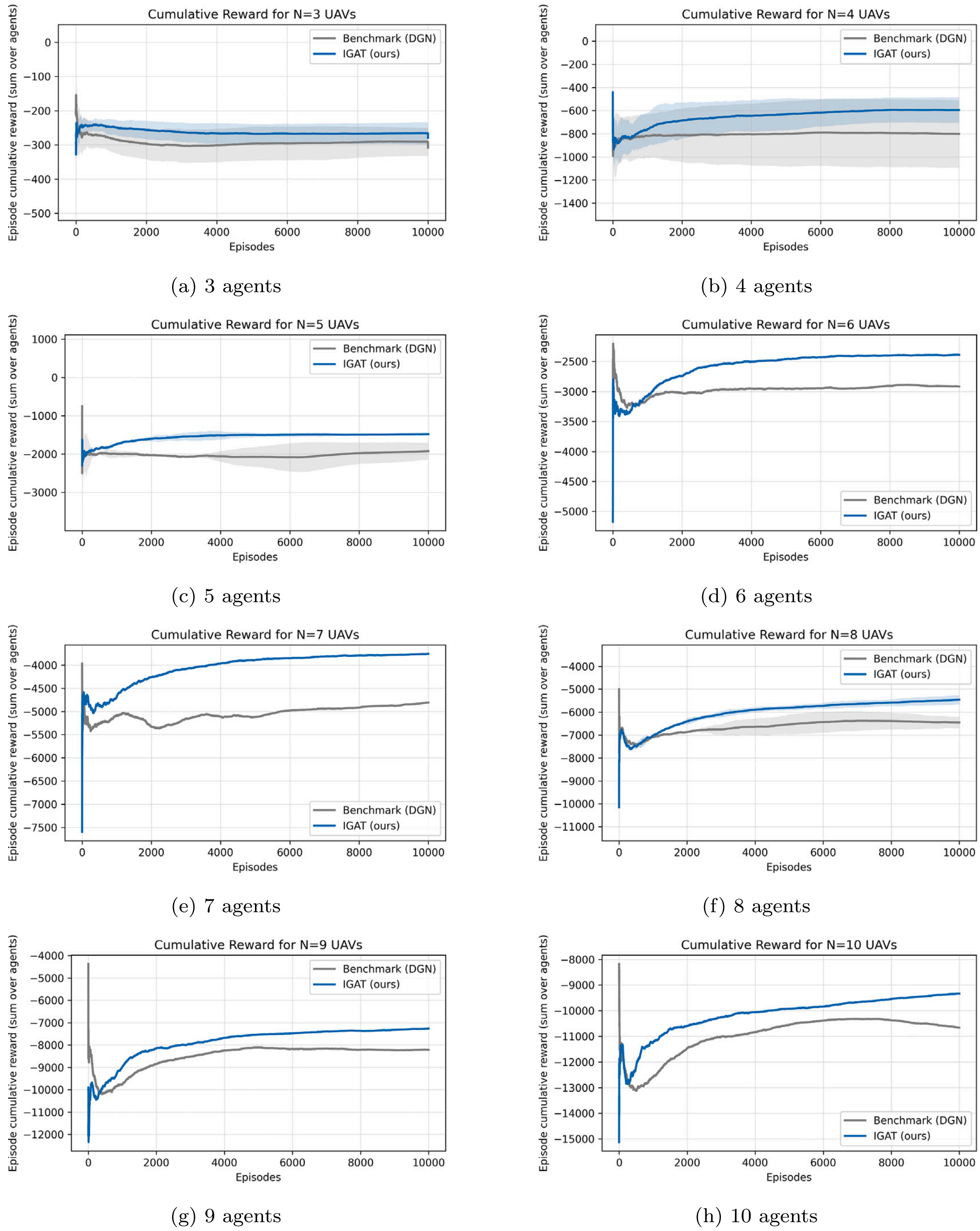


Fig. 2. Cumulative reward learning curves for 3 to 10 UAVs.

trade-offs by concentrating aggregation on neighbors that are important for safety, rather than making interactions more dense.

4.2.5. Comparison versus more graph-based MARL algorithms

We compare IGAT-MARL to four other graph-based MARL models in the same BlueSky environment, with the identical observation and action spaces, reward function, and replay buffer. The only difference is that we replace the graph approaches to see how the architecture affects the results. The primary benchmark is the conflict-masked QKV attention DGN (Graph-MARL) of Isufaj et al. [17]. We also compare a GRL-style multi-head dot-product attention module that scales aggregation by the conflict graph [37], an MS-GRL-inspired multiscale risk-fusion block that combines local attention with gated graph convolution [36], and a vanilla multi-head GAT baseline (MGAT) [27]. IGAT regularly finds a better balance between safety and efficiency in both medium and dense environments. It raises the total reward while minimizing the chance of losing separation and employing fewer active contact edges.

Table 2 shows the mean ± 95 percent confidence intervals for the previous 2000 episodes with N=5. IGAT has the finest overall balance. It achieves the highest cumulative reward (-1418 ± 48.8) and reduces separation-loss time to 461.6 ± 5.7. This indicates that the reward increased by 17.56% and the t_{loss} decreased by 10.52% compared to the benchmark (Table 3). IGAT also operates with fewer active edges (0.5245 ± 0.0352 vs. 0.9355 ± 0.0461), which demonstrates that greater coordination is achievable with a sparser interaction structure than with a denser one.

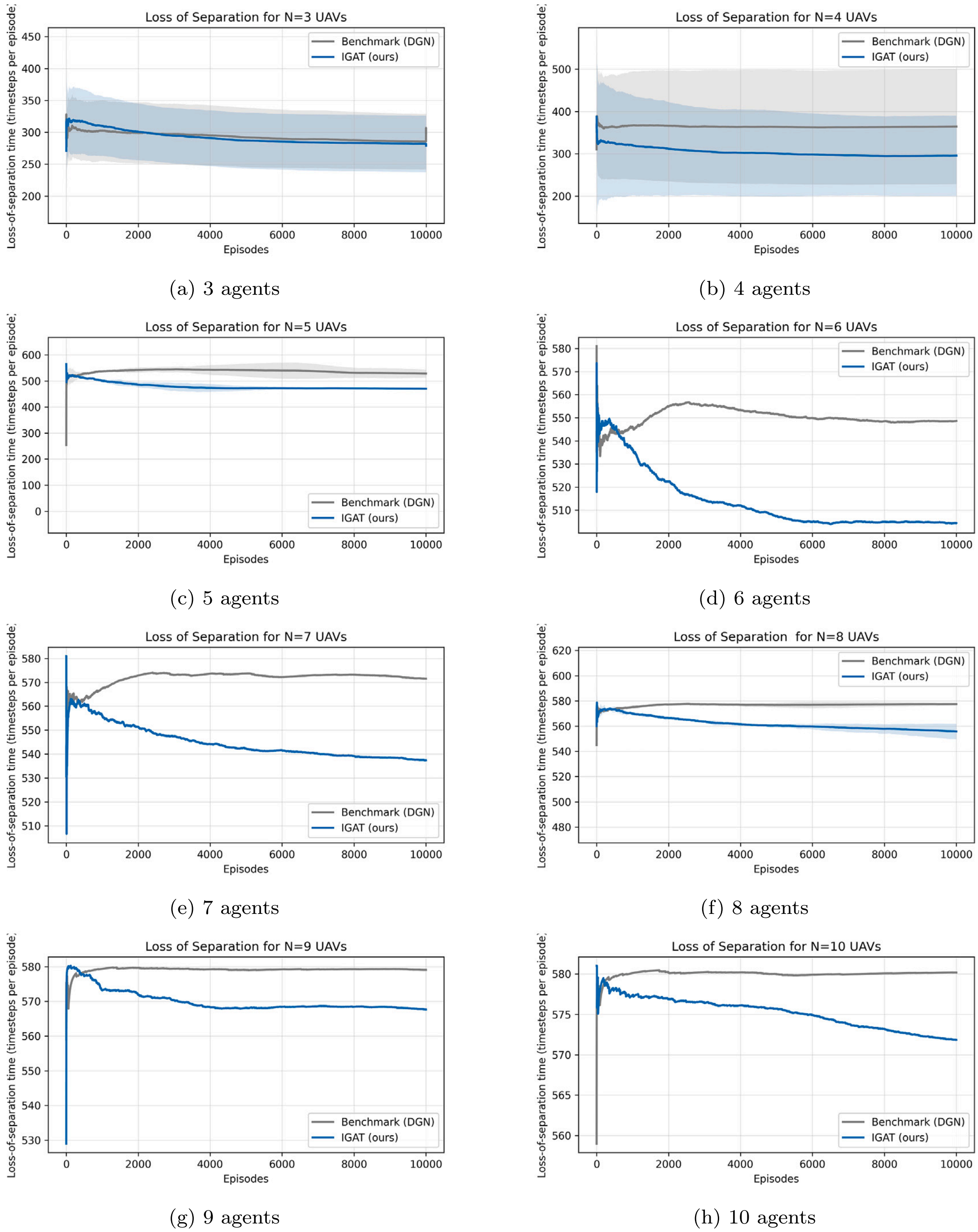


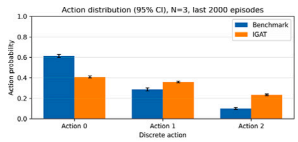
Fig. 3. Loss-of-separation duration (t_{loss}) during training for 3 to 10 UAVs.

Table 3 indicates how IGAT dominates other graph modules when the same simulator limitations and training process are used. IGAT increases cumulative reward by 17.56 percent and shortens loss-of-separation time by 10.52 percent compared to the benchmark DGN. It also uses 43.93 percent fewer active interaction edges. These gains show that making communication more dense doesn't lead to better performance; instead, it comes from learning a more selective conflict-centric interaction weighting over the same conflict-driven adjacency.

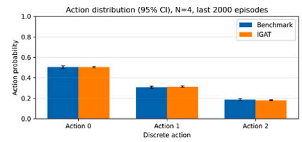
Significantly, the observed edge-count parameter, the number of edges, rises sub-quadratically with N in our scenarios, validating the assumption that our approach does not depend on densifying interactions as swarm size increases.

Figs. 5 show dense-swarm learning curves for all techniques that used the identical simulator, reward, and training procedure. The only difference is the graph aggregation module. IGAT achieves a larger total reward than the benchmark and other baselines, and it also keeps t_{loss} lower throughout training. The advantage persists, notwithstanding IGAT's utilisation of a conflict-driven sparse interaction framework. This supports the idea that focused coordination between safety-critical neighbors is better than passing messages around a lot of people at once. Figs. 5(a) and (b) show confidence bands that provide estimates of how uncertain the trajectories are.

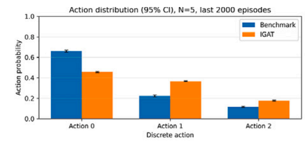
Table 4 shows performance with N = 10. IGAT attains a superior cumulative reward compared to the benchmark and the other baselines



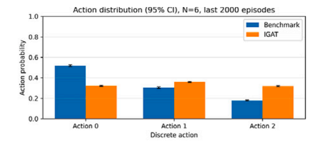
(a) 3 agents



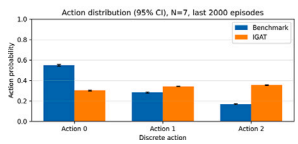
(b) 4 agents



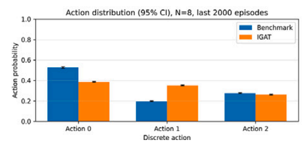
(c) 5 agents



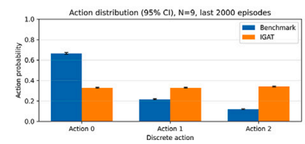
(d) 6 agents



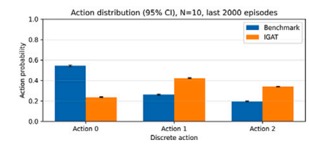
(e) 7 agents



(f) 8 agents



(g) 9 agents



(h) 10 agents

Fig. 4. Action distributions for benchmark and IGAT across swarm sizes (3 to 10). Bars show mean action probabilities over the last 1000 episodes and error bars show 95% confidence intervals.

Table 1
Comparison of the average number of network edges.

Agent Number	3	4	5	6	7	8	9	10
Benchmark (DGN)	0.183	2.138	1.654	1.474	2.742	3.573	3.892	4.671
IGAT (ours)	0.145	2.114	0.540	0.711	1.084	1.461	2.123	2.422

Table 2
Performance comparison for N=5 UAVs over the last 2000 training episodes (mean ± 95% CI).

Method	Acc. Reward ↑	t_{loss} ↓	Edges ↓
IGAT	−1418 ± 48.8	461.6 ± 5.7	0.5245 ± 0.0352
IGAT (no cur.)	−1649 ± 45.8	486.2 ± 4.58	0.5625 ± 0.0346
Benchmark (DGN)	−1719 ± 50.5	515.8 ± 4.57	0.9355 ± 0.0461
MGAT	−1782 ± 60.3	490.2 ± 6.31	0.8260 ± 0.0402
GRL	−1533 ± 49.2	484.8 ± 5.06	0.6275 ± 0.0393
MS-GRL	−2022 ± 44.8	505.7 ± 3.69	0.5500 ± 0.0338

Table 3
Relative improvement of IGAT vs baselines over the last 2000 episodes (N=5).

Baseline	Acc. Reward (%)	t_{loss} (%)	Edges (%)
IGAT (no curriculum/transfer)	+14.06	+5.07	+6.76
Benchmark (DGN)	+17.56	+10.52	+43.93
MGAT	+20.47	+5.84	+36.50
GRL	+7.54	+4.80	+16.41
MS-GRL	+29.88	+8.72	+4.64

Table 4
Comparison versus different graph-based MARL algorithms (N=10).

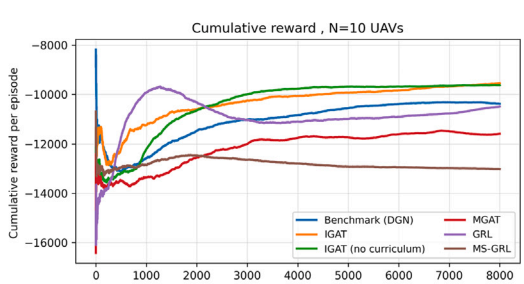
Method	Acc. Reward ↑	t_{loss} ↓	Edges ↓
IGAT (ours)	−9560.92	573.27	2.49
IGAT (no curriculum/transfer)	−9637.62	574.40	2.51
Benchmark	−10,384.75	580.05	4.37
MGAT	−11,599.46	578.47	3.89
GRL	−10,504.74	575.95	3.32
MS-GRL	−13,011.16	578.56	2.75

and fewer LoS and using fewer edges signify that gains are not acquired by randomly increasing the interaction density.

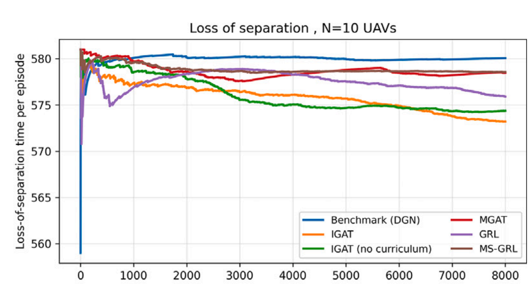
Table 5 shows how often actions occurred in all episodes. The benchmark is still highly biased toward $a = 0$, which indicates conservative behavior. IGAT makes the action distribution more even, which corresponds to a higher reward on a sparser graph. MGAT is moving toward a dominating turning action, which means that behavior is less likely to change (mode collapse risk). MS-GRL provides precise action probabilities, but it still struggles with incentives and LoS, suggesting that having balanced actions isn't sufficient without a meaningful way to assess the value of conflict.

4.2.6. *Curriculum and transfer learning*

Table 6 compares the proposed IGAT trained with curriculum learning and inter-stage transfer (Curr + TL) against the same IGAT ar-



(a) Cumulative reward.



(b) Loss of separation.

Fig. 5. Comparison with different graph MARL algorithms: Cumulative rewards and loss of separation for 10 UAV collision–avoidance scenarios with 95% confidence.

Table 5
Action selection bias over the 8000 episodes.

Method	Action 0	Action 1	Action 2
IGAT (ours)	0.302	0.350	0.348
IGAT (no curriculum/transfer)	0.319	0.351	0.329
Benchmark	0.486	0.282	0.232
MGAT	0.423	0.261	0.316
GRL	0.335	0.448	0.217
MS-GRL	0.333	0.334	0.333

Table 7
IGAT ablation study (5 UAVs, 10,000 training episodes). Reported values are mean ± 95% CI over all 10,000 episodes.

Method	Acc. Reward ↑	t_{loss} ↓	Edges ↓
IGAT (Full-2Layer)	−1467.665 ± 21.820	470.128 ± 2.412	0.541 ± 0.016
IGAT (1Layer)	−1721.090 ± 23.264	500.413 ± 2.018	0.808 ± 0.021
IGAT (1IGAT in 1Layer)	−2095.212 ± 30.696	532.132 ± 1.807	1.569 ± 0.031

Table 6
Ablation of curriculum + transfer learning (Curr + TL) for IGAT across $N = 4$–10 UAVs. Values are mean ± 95% CI over the last 200 episodes within the first 2000 training episodes.

N	Method	Acc. Reward ↑	t_{loss} ↓
4	IGAT (Curr + TL)	−538.36 ± 57.31	255.42 ± 8.47
	IGAT (No Curr/TL)	−818.63 ± 81.55	411.21 ± 17.36
5	IGAT (Curr + TL)	−1310.02 ± 155.70	444.51 ± 20.98
	IGAT (No Curr/TL)	−1756.57 ± 142.58	494.64 ± 14.12
6	IGAT (Curr + TL)	−2323.75 ± 222.81	504.65 ± 13.59
	IGAT (No Curr/TL)	−2489.73 ± 198.54	514.04 ± 12.14
7	IGAT (Curr + TL)	−3517.08 ± 274.43	529.26 ± 10.75
	IGAT (No Curr/TL)	−3700.69 ± 287.85	529.74 ± 10.70
8	IGAT (Curr + TL)	−4764.74 ± 385.36	543.34 ± 8.91
	IGAT (No Curr/TL)	−5649.73 ± 356.12	555.05 ± 8.07
9	IGAT (Curr + TL)	−6691.55 ± 436.19	561.60 ± 6.51
	IGAT (No Curr/TL)	−7510.44 ± 351.25	570.22 ± 4.83
10	IGAT (Curr + TL)	−8949.81 ± 598.01	567.88 ± 6.07
	IGAT (No Curr/TL)	−9144.34 ± 437.51	570.23 ± 4.19

chitecture trained directly without curriculum/transfer (No Curr/TL), evaluated over the first 2000 episodes (statistics computed on the last 200 episodes within this window). For $N = 4$ to 10, Curr + TL always increases the early-stage return (i.e., less negative reward), which means that the collision-avoidance strategy is improving more quickly. The most significant improvements occur in smaller to medium-sized swarms, where exploration is most unstable. When $N = 4$, Curr + TL raises the reward by 280.27 (about 34 percent reduction in reward penalty magnitude) and decreases the loss-of-separation time t_{loss} by 155.79 (about 38 percent). For $N = 5$ and $N = 8$, Curr + TL exhibits clear improvements in both safety and reward. This illustrates that it works well to start with smaller swarms and then progress to larger ones. In dense MARL, Curr + TL is projected to have an early-stage advantage because adding more agents makes the environment less stable and increases the likelihood that naive exploration will lead to dangerous paths and inaccurate credit assignment. Curriculum learning simplifies this by first learning reliable ways to resolve conflicts in smaller swarms. Transfer learning, on the other hand, starts the subsequent phases with parameters that already encode how to prioritize neighbors and estimate their worth. Curr + TL reduces the amount of "random" risky exploration that has to occur in larger-N settings. This leads to reduced t_{loss} and better returns in the first 2000 episodes. The steady gains in reward and t_{loss} demonstrate that Curr + TL primarily helps policies develop more quickly and safely during the most unstable part of training.

4.2.7. Ablation on attention depth and refinement

To assess the value of IGAT architecture, we conduct ablations that reduce the depth of the attention layers while maintaining a constant environment, reward, and training configuration. Tables 7 and Figs. 6 show that the full IGAT always increases reward and lowers t_{loss}, and it also needs fewer edges than reduced-depth versions. This supports the idea that IGAT's stacked refinement (which includes "double attention" with normalization and residual stabilization) helps conflict-centric coordination when the interaction topology changes over time.

In the N=10 configuration, IGAT without curriculum/transfer also beats all the baselines when all the provided metrics are taken into account. As shown in Table 4, it achieves the highest cumulative reward while also maintaining a competitive loss-of-separation (LoS) and the lowest average interaction edges. This demonstrates that the IGAT backbone itself remains effective in dense traffic even without staged training. This implies that curriculum/transfer primarily makes training more stable and aids convergence, while the most significant performance improvements in dense swarms stem from IGAT's conflict-centered attention and selective aggregation.

From an operational standpoint, the findings indicate that conflict-oriented sparse interaction graphs may enhance safety and efficiency without necessitating dense communication. This is important for areas with a lot of UAV activity where bandwidth and onboard computing power are limited, and where needless contacts with neighbors might create noise and slow down decision making. In real deployments, there may be sensing uncertainty, communication delays, and latency limits that could make it harder to detect and coordinate conflicts. In addition to extending the action area to include vertical or speed movements, these elements are not specifically covered here but are logical directions for future expansion.

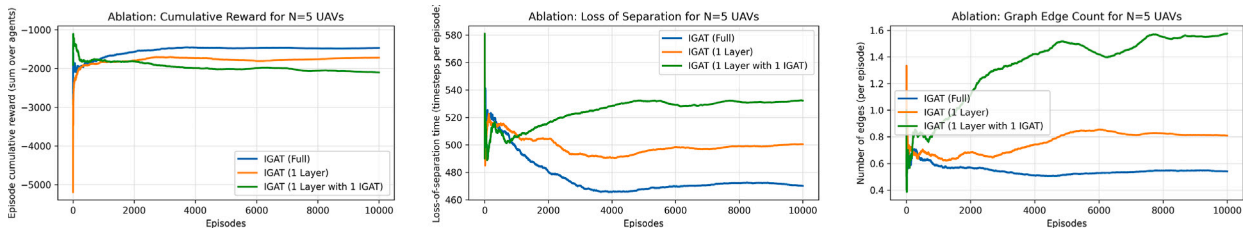
5. Discussion

According to the results, IGAT offers a better overall trade-off than the benchmarks because it produces fewer active interaction edges, a greater cumulative reward, and a lower separation loss. Additionally, IGAT reduces action-selection bias, making it easier for the policy to adapt and avoid getting stuck on the optimal course of action. These numbers show that the improvement is not due to greater communication but to more targeted coordination.

The design of IGAT aligns with modifications that make it safer and work better. The adjacency based on conflict creates a sparse interaction graph, and the attention mechanism learns to focus on the neighbors that are most important for safety among the active conflict pairs. The value network may focus on interactions that provide it with relevant information and disregard those that do not. This may make it more stable when there are many agents that are not stationary. The decreased edge count should not be taken as a direct signal of reduced conflict potential; instead, it should be viewed as a manifestation of simpler graphs and message transmission under the same conflict parameters.

Parameter transfer training in the curriculum also makes it easier to scale things up. The technique could employ coordination patterns and value estimates learned earlier by starting with smaller swarms and gradually adding more UAVs. As the space of interactions becomes more complex, the system becomes more stable and convergent. Also, our scenario generation ensures that agents must deal with genuine disputes throughout training. This saves students from having to take lengthy, boring classes and teaches them how to handle problems consistently.

The findings show that focusing on conflict-relevant neighbors may help with coordination while keeping interaction low. This is important in crowded areas where computing and communication resources are limited. We don't explicitly discuss these features here, and we need to undertake more research on perception that considers ambiguity, delayed information sharing, and wider action regions (such as managing speed and altitude).



(a) **N=5:** Cumulative reward.

(b) **N=5:** Loss-of-separation time (t_{loss}).

(c) **N=5:** Number of edges.

Fig. 6. Ablation study of IGAT; showing cumulative reward, loss-of-separation duration, and average number of active interaction edges.

6. Conclusion and future work

In this paper, we presented a multi-agent learning system for UAV collision avoidance, leveraging attention graphs to assist UAVs prioritize adjacent UAVs and make more educated navigation choices. Simulation findings reveal that the suggested strategy efficiently decreases collision risk, even in heavily inhabited multi-UAV airspaces.

For future study, we aim to strengthen the system by including a broader variety of collision avoidance actions, including height and speed modifications, to increase flexibility in complicated surroundings. Additionally, although this research focused primarily on UAV-to-UAV collisions, future work will entail accounting for additional dynamic and static impediments to generate more realistic and challenging situations. Lastly, our present tests were conducted on fixed-wing UAVs; extending testing to other UAV types, such as quadrotors, would further demonstrate the algorithm's adaptability and applicability across diverse platforms.

CRediT authorship contribution statement

Mohammad Reza Rezaee: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nor Asilah Wati Abdul Hamid:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition. **Masnida Hussin:** Supervision. **Zuriati Ahmad Zukarnain:** Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered potential competing interests:

Nor Asilah Wati Abdul Hamid reports that financial support was provided by Air Force Office of Scientific Research. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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